

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL AS CHEMISTRY (9620)

Unit 1: Inorganic 1 and Physical 1

Wednesday 8 May 2024 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do **not** write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 70.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
TOTAL	



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ANSWER IN THE SPACES PROVIDED**



Answer **all** questions in the spaces provided.

0 1

This question is about zinc.

0 1 . 1

Complete the electron configuration for an atom of zinc.

[1 mark]

$1s^2$ _____

0 1 . 2

State the number of full 3d orbitals in an ion of Zn^{2+}

[1 mark]

0 1 . 3

An atom of a different element has two more protons and three more neutrons than an atom of ^{65}Zn

Give the symbol and the mass number of this atom.

[2 marks]

0 1 . 4

State why zinc is a good conductor of electricity.

[1 mark]

A sample of zinc contains three isotopes, ^{64}Zn , ^{66}Zn and ^{68}Zn

0 1 . 5

Explain why these isotopes have the same chemical reactions.

[1 mark]

Question 1 continues on the next page

Turn over ►



0 1 . 6

The sample is ionised by electron impact and then analysed using time of flight (TOF) mass spectrometry.

The ions are accelerated to a kinetic energy (KE) of $1.25 \times 10^{-16} \text{ J}$
They travel down a 0.650 m flight tube.

$$KE = \frac{1}{2}mv^2$$

m = mass / kg

v = velocity / ms^{-1}

Calculate the mass, in kg, of a $^{64}\text{Zn}^+$ ion.

Calculate the time, in s, for a $^{64}\text{Zn}^+$ ion to travel through the flight tube.

The Avogadro constant, $L = 6.022 \times 10^{23} \text{ mol}^{-1}$

[4 marks]

Mass of a $^{64}\text{Zn}^+$ ion _____ kg

Time to travel through the flight tube _____ s



0 1 . 7 Table 1 shows the data from the mass spectrum of the sample of zinc.

Table 1

$\frac{m}{z}$	Relative intensity
64	2.8
66	1.6
68	1.0

Calculate the relative atomic mass (A_r) of zinc.
Give your answer to 1 decimal place.

[2 marks]

A_r _____

12

Turn over for the next question

Turn over ►



0 2

Sodium hydrogencarbonate reacts with sulfuric acid.



A student analyses some impure sodium hydrogencarbonate in a titration experiment. The impurity is unreactive.

Step 1 The student weighs a 2.21 g sample of impure sodium hydrogencarbonate and dissolves the solid in distilled water in a beaker.

Step 2 The student transfers the solution, with washings, to a 100 cm³ volumetric flask and makes up to the mark with distilled water.

Step 3 The student transfers 25.0 cm³ of the solution into a conical flask using a pipette.

Step 4 The student adds a few drops of an indicator to the conical flask and adds 0.100 mol dm⁻³ sulfuric acid from a burette until the indicator changes colour.

Step 5 The student repeats **Steps 3** and **4** twice.

Table 2 shows the results of the titrations.

Table 2

	1	2	3
Final volume / cm³	35.40	40.25	37.65
Initial volume / cm³	3.00	8.00	5.45
Titre / cm³	32.40	32.25	32.20

0 2 . 1

Use suitable values from **Table 2** to calculate the mean titre.

[1 mark]

Mean titre _____ cm³



0 2 . 2

Use your answer to Question **02.1** to calculate the percentage, by mass, of sodium hydrogencarbonate in the sample.
Give your answer to 3 significant figures.

[5 marks]

_____ %

0 2 . 3

Describe a method to determine the accurately known mass of solid added to the beaker in **Step 1**.

[2 marks]

0 2 . 4

State why the student used a conical flask rather than a beaker for the titration in **Step 4**.

[1 mark]

Question 2 continues on the next page

Turn over ►

0 2 . 5

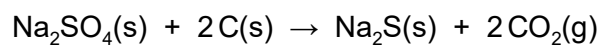
Sodium sulfate is a solid with a high melting point.

Explain why the melting point of sodium sulfate is high.

[2 marks]

0 2 . 6

At high temperatures sodium sulfate reacts with carbon to form sodium sulfide and carbon dioxide.



3.28 g of sodium sulfate are reacted with an excess of carbon.

4850 cm³ of carbon dioxide are formed at a pressure of 100 kPa

Calculate the temperature, in °C, of the carbon dioxide.

The gas constant, $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$

[6 marks]

_____ °C

17



0 3

This question is about Period 2 elements.

0 3 . 1

Table 3 shows the first ionisation energies of some Period 2 elements.**Table 3**

Element	First ionisation energy / kJ mol^{-1}
Lithium	519
Beryllium	900
Boron	To be predicted
Carbon	1088
Nitrogen	1402

Predict the value of the first ionisation energy of boron.

Explain your answer in terms of atomic structure.

[3 marks]First ionisation energy of boron _____ kJ mol^{-1}

Explanation _____

Question 3 continues on the next page**Turn over ►**

0 3 . 2 Table 4 shows the electronegativity values for some of the elements in Period 2.

Table 4

	Li	Be	B	C	N	O	F
Electronegativity	1.0	1.5	2.0	2.5	3.0	3.5	4.0

Suggest why electronegativity values increase across Period 2.

[2 marks]

0 3 . 3 Draw the shape of a molecule of boron trifluoride, BF_3
Include any lone pairs, if any, that influence the shape.

Explain why a molecule of boron trifluoride is non-polar, although a B—F bond is polar.

[3 marks]

Shape

Explanation _____



0	3	.	4
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A boron trifluoride molecule reacts with a fluoride ion to form a BF_4^- ion.

Name the type of bond formed when BF_3 reacts with a fluoride ion.

Explain how this bond is formed.

[2 marks]

Name of bond _____

Explanation _____

10

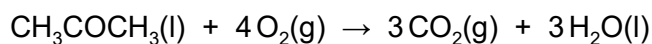
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0 4

This question is about the standard enthalpy of combustion of propanone, CH_3COCH_3



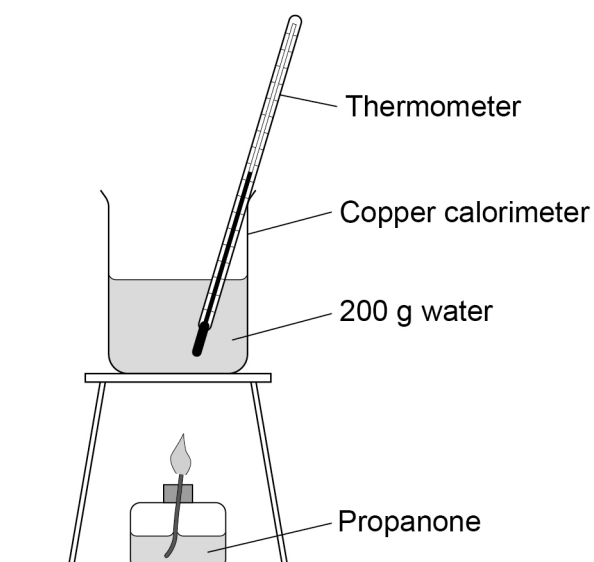
0 4 . 1

Define standard enthalpy of combustion.

[2 marks]

0 4 . 2

In a calorimetry experiment, 1.027 g of propanone are burned in oxygen. The heat given out is used to increase the temperature of 200 g of water from 19 °C to 42 °C



Calculate a value, in kJ mol^{-1} , for the enthalpy of combustion of propanone. The specific heat capacity of water is $4.18 \text{ J K}^{-1} \text{ g}^{-1}$

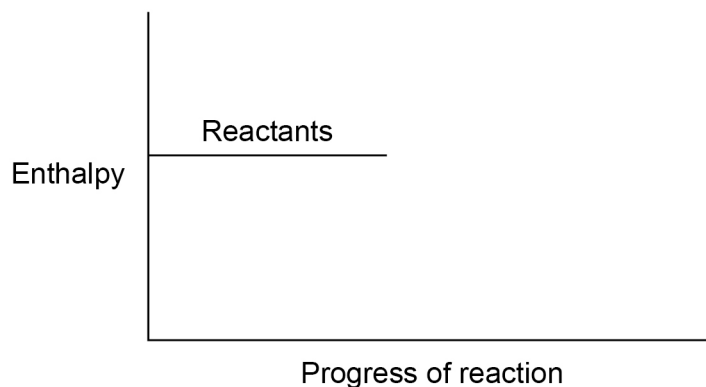
[4 marks]

Enthalpy of combustion _____ kJ mol^{-1}



0 4 . 3 Complete the enthalpy level diagram for the combustion of propanone.

[1 mark]



0 4 . 4 Suggest why a copper calorimeter is used rather than a glass beaker.

[1 mark]

In a data book, the value for the enthalpy of combustion for propanone is $-1792 \text{ kJ mol}^{-1}$

0 4 . 5 Give one reason, other than heat loss, why the value obtained in Question **04.2** is different from the value given in the data book.

[1 mark]

Question 4 continues on the next page

Turn over ►



0 4 . 6

The standard enthalpy of combustion of propanone can be used to calculate the standard enthalpy of formation of propanone.

The equation for the standard enthalpy of formation of propanone is

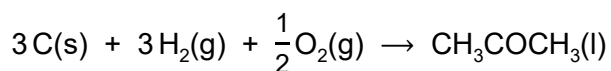


Table 5 shows some standard enthalpy of combustion data.

Table 5

	C(s)	H₂(g)	CH₃COCH₃(l)
$\Delta_c H^\ominus / \text{kJ mol}^{-1}$	−394	−286	−1792

Use the data in **Table 5** to calculate the standard enthalpy of formation, in kJ mol^{-1} , of propanone.

[2 marks]

Standard enthalpy of formation _____ kJ mol^{-1}

11



Turn over for the next question

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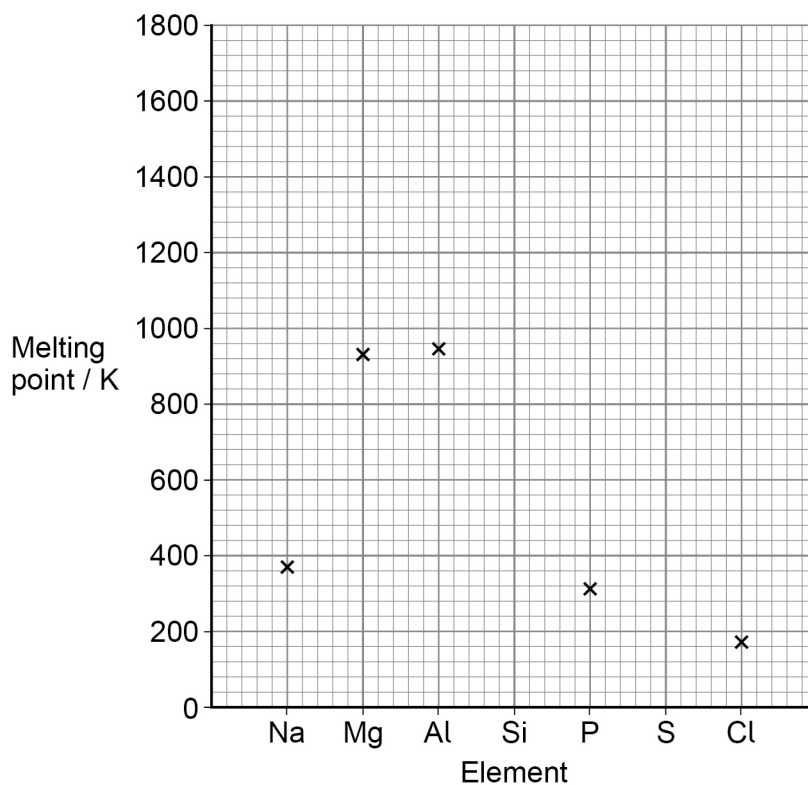
0 5

This question is about the Period 3 elements, sodium to chlorine, and their compounds.

0 5**1**

Figure 1 shows the melting points of some of the elements in Period 3.

Figure 1



Draw crosses on **Figure 1** to mark the melting points for silicon **and** sulfur.

[2 marks]

0 5**2**

In terms of structure and bonding, explain why magnesium has a higher melting point than sodium.

[2 marks]



0 5 . 3

In terms of structure and bonding, explain why phosphorus has a higher melting point than chlorine.

[2 marks]

0 5 . 4

Which of these elements has the largest atomic radius?

Tick (✓) **one** box.

[1 mark]

sodium

☐

silicon

☐

sulfur

☐

0 5 . 5

Chlorine reacts with cold, dilute aqueous sodium hydroxide.

Write an equation for the reaction.

[1 mark]

0 5 . 6

Chlorine reacts with an aqueous sodium halide to form a black solid in a dark brown solution.

Identify the sodium halide.

Write an ionic equation for the reaction.

[2 marks]

Identity of sodium halide _____

Ionic equation

Turn over ►

0	5	.	7
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Sodium chloride dissolves in water to form a colourless solution.

A student adds dilute acid and another reagent to show that the colourless solution contains chloride ions.

Give the formula of this other reagent.

Give one observation the student would make.

[2 marks]

Formula of reagent _____

Observation _____

0	5	.	8
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Solid sodium bromide reacts with concentrated sulfuric acid in a redox reaction.

Write an equation for this reaction.

State **one** observation.

[2 marks]

Equation

Observation _____

14



0 6

This question is about redox reactions containing manganese compounds.
 MnO_2 reacts, in acidic conditions, with iodide ions to form iodine.

0 6 . 1

Write a half-equation to show the conversion of iodide ions into iodine.

[1 mark]

0 6 . 2

Write a half-equation to show the conversion of MnO_2 , in acidic conditions, into Mn^{2+}

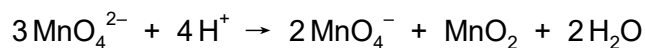
[1 mark]

0 6 . 3

Write an overall ionic equation for the reaction of MnO_2 , in acidic conditions, with iodide ions.

[1 mark]

MnO_4^{2-} reacts, in acidic conditions, as shown in the equation.

**0 6 . 4**

Give the oxidation state of manganese in MnO_4^{2-} **and** in MnO_4^-

[1 mark]

Oxidation state of Mn in MnO_4^{2-} _____

Oxidation state of Mn in MnO_4^- _____

Question 6 continues on the next page

Turn over ►

0	6	.	5
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In the extraction of manganese from Mn_3O_4 , aluminium acts as a reducing agent and forms Al_2O_3

State, in terms of electrons, the meaning of reducing agent.

Write an equation for the overall reaction.

[2 marks]

Meaning of reducing agent _____

Equation

6

END OF QUESTIONS



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[illegible]

2 4



2 4 6 X C H O 1